Network Security Formula Sheet (Chapter 8)

1. Cryptographic Operations

Symmetric Key Cryptography

Alice and Bob share a single secret key, K_S.
- **Encryption:** $c = K_S(m)$
- **Decryption:** $m = K_S(c)$

Public Key (Asymmetric) Cryptography for Confidentiality

Bob has a public key K_B^+ and a private key K_B^-.

$$m = K_B^-(K_B^+(m))$$

Encryption (by Sender/Alice): Encrypt the message 'm' using the receiver's **public key**. The ciphertext is $c = K_B^+(m)$.
Decryption (by Receiver/Bob): Decrypt the ciphertext 'c' using the receiver's **private key**. The message is $m = K_B^-(c)$.

2. The RSA Algorithm

The most widely used public key algorithm. Based on modular exponentiation.

RSA Key Generation

1. Choose two large prime numbers, p and q.
2. Compute modulus $n = p \cdot q$.
3. Compute Carmichael's totient function (or Euler's totient for original RSA) $z = (p-1)(q-1)$.
4. Choose public exponent e such that $e < n$ and e is relatively prime to z.
5. Compute private exponent d such that $(e \cdot d) \pmod{z} = 1$. This is the modular multiplicative inverse.
6. **Public Key:** (n, e)
7. **Private Key:** (n, d)

RSA Encryption and Decryption

Encryption: To encrypt a message 'm' (represented as a number), compute the ciphertext 'c':

$$c = m^e \pmod{n}$$

Decryption: To decrypt a ciphertext 'c', compute the original message 'm':

$$m = c^d \pmod{n}$$

RSA's Dual Property (for Digital Signatures)

The order of applying the keys can be reversed, which is the mathematical foundation of digital signatures.

$$K_B^-(K_B^+(m)) = m = K_B^+(K_B^-(m))$$

The first term is encryption for confidentiality. The second term, applying the private key first, is the basis for signing.

3. Digital Signatures and Message Integrity

Digital Signature Creation (by Sender/Bob)

1. Compute the cryptographic hash of the message: $h = H(m)$.
2. Create the signature 'S' by encrypting the hash with the sender's **private key**:

$$S = K_B^-(h) = K_B^-(H(m))$$

3. The sender sends the pair (m, S).

Digital Signature Verification (by Receiver/Alice)

1. Compute the hash of the received message: $h' = H(m)$.
2. Decrypt the received signature 'S' using the sender's **public key** to recover the original hash:

$$h = K_B^+(S)$$

3. **Check:** If $h' = h$, the signature is authentic and the message has integrity. Otherwise, it is invalid.

4. Modular Arithmetic Rules

These rules are the foundation for why RSA works.

$$[(a \pmod n) + (b \pmod n)] \pmod n = (a + b) \pmod n$$

$$[(a \pmod n) \cdot (b \pmod n)] \pmod n = (a \cdot b) \pmod n$$

$$(a \pmod n)^d \pmod n = a^d \pmod n$$